

Augmenting Movie Test Screenings: A Physiological Data Processing Pipeline for Predicting Film Perception and Profitability

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Summary

Movie test screenings allow film studios to gauge audience reactions before the final product is released and to encourage more informed post-production decisions that have serious implications regarding future commercial outcomes. Current practices solely rely on self-reports after viewing experience, which are critiqued in psychological and consumer-neuroscience literature. This thesis investigates the possibilities of augmenting test screening processes within the film industry by adding data from wearable physiological reaction measurement devices that are processed through a scalable pipeline, with the goal of predicting and better understanding the dynamics of movie perception (operationalized by IMDb rating as an outcome variable) and profitability (operationalized by Word-of-Mouth multiplier as an outcome variable).

The research is framed by two research questions addressing methodological (RQ1) and empirical (RQ2) dimensions, using the Emognition dataset.

RQ1 is answered by the creation of a data processing pipeline that is scalable, requires less resources in comparison to the original raw data representation, and is usable in a real-life scenario. The pipeline includes participant-level, multimodal feature extraction, participant baseline correction, movie-level aggregation, addition of manually collected metadata on scenes and movies, and modelling/analysis using regression, classification and deep learning models.

RQ2 is addressed by the direct application of this pipeline on the Emognition dataset. Due to low number of real observations, copula-based data simulation is performed ($n=50$ for ML, $n=10010$ for DL). In addition, two types of evaluation setups are proposed: Leave-One-Out CV for evaluating models on real data only, and k-fold CV on models using synthetic data for training as well. Machine learning models are selected to represent different biases and failure conditions with the goal of capturing as many different signals from the data as possible.

Across model types and data scales, perception is predicted approximately twice as reliably as profitability (best IMDb $R^2 = 0,76$ and WOM $R^2 = 0,33$). Spearman ρ exceeds Pearson r , which exceeds R^2 in every regression configuration. At the strongest setting, the pipeline performs relative IMDb ordering at $\rho = 0,82$ from synthetic-only training. Commercial failure is detected more reliably than commercial success: five of six classifiers reach $AUC = 0,81$ on flop detection, while hit detection peaks at $AUC = 0.75$ with most classifiers unable to identify any positive example. Physiology alone preserves 88% (random forest model) of predictive performance achievable with the full variables set, indicating that self-report contributes relatively little predictive value in this configuration. Cross-method convergence to identical numerical results, consistency across data scales, and a low R^2 increase to 200x training data augmentation indicate that the pipeline operates at a ceiling set by the Emognition dataset size of ten real films rather than by methodological capacity.

The findings are bounded by the data bottleneck due to low number of real observations, using few-minute clips rather than full films as stimuli, and by the specific Word-Of-Mouth and IMDb constructs used to operationalise profitability and perception. The patterns observed are signals from these specific movie clips and this specific stimulus setting. Generalization to all movies and whole populations of target segments is outside the scope of research. Replacement of self-reports is not claimed either.

The thesis achieved results which support the proposition that physiological data could serve as complementary to self-reports in test screening scenarios and create business value. With explicit limitations, contribution is made on a methodological level by providing an applicable and feasible pipeline for business usage, and on an empirical basis by identifying key patterns and use cases which may help film studios make better business decisions.

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1. Introduction

Motion pictures have been part of history for well over a century now. With technological advancement a new art form was born that is able to convey stories, feelings and ideas through countless different lenses in an accessible way. This audio-visual way of artistic expression, often serving as an investigation and critique of our world and the human condition led to many timeless masterpieces over the years. The depth and influence of film and cinema is undeniable and spreads through numerous endeavours in the areas of technology, art, culture or business. Each movie is a subjective experience that becomes something different, observed through the perspective of the viewer.

Cinema has come a long way from the early days of silent films and monochrome productions. In the modern era, it became a huge and influential industry, rising through the fascinating merge of art and business. Giant production studios dominate the global market, releasing large-scale projects and bringing multitudes of people into the movie theatres. With the rise of streaming distribution, the volume of film production and the size of global audiences are both at historic highs.

Therefore, the commercial nature and aspect of this form of art cannot be neglected. There is tremendous business value to be found in addition to the artistic and cultural significance. High-budget blockbusters cost hundreds of millions of US dollars, making them large-scale investments for all parties involved. There is a large amount of risk involved, which businesses try to gain as much of an understanding into as possible. Studios closely follow and research trends and various influential factors in terms of movie profitability, sometimes even going against artistic freedom of expression and deeper cultural value.

Due to the nature of movies being subjective audio-visual experiences, understanding of customers is an intricate issue. There are countless research areas and factors intertwined, where gaining full understanding is practically an impossible task. Nevertheless, gaining insight into what experiences people go through during moviegoing or simply having a movie night as good as possible has crucial implications in both psychological and business areas. This directly leads to changes in critical acclaim, contribution towards societal values, overall public image and commercial success.

Countries with advanced film industries and big studios often use test screenings in the post-production phase, in order to gauge audience reactions via self-reports after the viewing experience. This thesis investigates whether physiological measurement (captured via wearable devices during clip viewing) can complement traditional self-report instruments at this evaluation stage, and whether a machine-learning pipeline applied to such recordings can produce defensible predictions of audience response.

1.1 Problem statement

Movie test screenings have been around since the very early days of the film industry. This is the process of a carefully selected group of people (sampled so that they adequately cover the target audience population) participating in a movie screening before the product is finalized and released. Afterwards, based on people's feedback in form of surveys and questionnaires,

these big production studios evaluate the movie perception in terms of various factors including overall reception, understanding of the narrative and playability (Goetz, 2026).

The results can greatly influence creative decisions made in the post-production phase of a film's life cycle. There are many examples that highlight the significance of test screenings, even regarding some of the biggest, most successful and well-known movies of all time. Martin Scorsese's iconic gangster epic, *Goodfellas*, had significant scenes changed in terms of pacing and the amount of violence due to unfavourable reactions from the test audience. The first version of James Cameron's iconic blockbuster, *Titanic*, was also edited and cut in accordance with initial reactions (Obias, 2016). Even though it is crucial to keep the original idea and vision of the artists involved, such smaller changes can be vital turning points in the film's afterlife on the market in terms of both turnover, critical acclaim and opinion of the general public.

Thus, it is very important in a business context to understand the dynamics and effects surrounding a test screening. Current methods on such occasions only go so far as to ask the audience themselves on their opinion of the films. Forms vary, some use more complex and scientifically established surveys, questionnaires, while some let the customers express their thoughts in a broader sense in whole sentences. In practice, however, no matter how well the data collection method is constructed, it has serious limitations, making it oftentimes ineffective and misleading. Psychological research shows that people are often unable to introspect on their mental processes accurately (Nisbett, 1977). These self-reports therefore come with the risk of containing mere rationalisations, not the actual emotional impact of a film product. In addition, more modern marketing research shed a light on how a big proportion of our buying decisions are made subconsciously (Zaltman, 2003). A significant measurement gap therefore exists within the film industry: current practices of test-screening fail to capture some of the signals from audience response, which leads to less informed high-impact business decisions.

1.2 Purpose

The main proposal of this thesis is to investigate the feasibility and usefulness of the inclusion of physiological data at test screenings. It is not claimed that self-reports should be completely replaced from the beginning, however the extent to which they matter is tested, evaluated and reflected upon. Comprehensive collection of physiological data may prove to be a significant addition, serving as a complementary feature to add validity and capture signals from audience reactions that self-reports would otherwise be unable to. The Emognition dataset (Saganowski S. K., 2022), later used in this paper, proposes a data collection methodology that combines three wearable devices, covering many aspects of emotional recognition.

By the introduction of such additions, movie studios may be able to have a better understanding of the actual effects of their films on customers. This leads to more informed decisions, influencing the overall reception of the movie, with corresponding implications of commercial outcome and success. Admittedly, getting such devices and setting up test screenings of this kind on a big scale is much more expensive in comparison to standard practices. However, it is more than feasible for large studios in the entertainment industry, especially if the profitability of big-screen productions is in question.

There are numerous areas touched upon while researching this topic including business, psychology, art and even sociology. It is a very complicated topic but as a result, a deeper understanding may be gained regarding the handling, preparation and analysis of such datasets containing physiological signals, investigation of human reactions to work of art, emotions, and various business aspects. The main focus is on creating business value by improving efficiency and understanding, but there may be various other aspects creating room for further research.

1.3 Research questions

Broadly speaking, the main question this thesis aims to answer is the following: *Can physiological reaction measurement through a scalable pipeline, serve as a complement to self-report surveys in film test-screening practices for predicting audience perception and profitability?* In order to address this question, two problems need to be addressed, and for both, a research question is constructed.

Firstly, there must be a reproducible and efficient pipeline to handle the physiological data collected, suitable for model training and flexible in terms of future research. It is a crucial challenge to face in order to help the actual implementation of such an addition to test screenings. Scalability is key, not only theoretically due to the nature of machine learning and deep learning models, but practically as well. Huge studios that would most likely benefit from the implementation of such practices have multiple screenings with hundreds of people. Enormous amounts of multi-dimensional data is generated that is not model-ready. Therefore, the first research question is:

RQ1: How can a scalable and theory-driven pipeline be designed to extract and analyse predictive signals of audience response to films from physiological data?

Having an established pipeline carries value in itself. However, in order to actually produce insights and results from the data, the pipeline needs to be applied. In this research, the Emognition dataset will be used. While it has a set of limitations, the dataset contains multimodal physiological data of participants watching movie scenes, making it a suitable choice. The main areas of interest are perception and profitability. Both are very complex and delimitations are introduced to make the research feasible. Outcome variables are also computed and adjusted to fit the scope of this research. Thus, the second research question is:

RQ2: What insights into the relationship between physiological reactions to film clips and the corresponding films' audience perception and profitability can be gained by applying this pipeline to the Emognition dataset?

By answering these two research questions, the contribution of the thesis is defined in both a methodological and empirical dimension.

1.4 Delimitation

The research areas this thesis is written within and in relation to are extremely complex with countless influential factors. Defining a reasonable scope and delimitations are crucial when dealing with movies as products of business endeavours, psychological behaviour of customers and physiological reactions of human beings. These delimitations must be defined clearly and reflected upon throughout the research process.

First of all, some concept-level delimitations are introduced to narrow down the research focus and create a feasible project. The Emognition dataset is the sole source of physiological data. This comes with restrictions regarding the dataset: participants are young adult Europeans, a demographic that does not exhaustively represent the cinema-going population. The stimulus set does not contain full movies, only 2-minute-long clips. The selected movies in question are also well-known; their success varies, but none of them are complete financial failures.

Concepts such as profitability and perception are operationalised through measurable outcome variables (IMDb rating for perception and the word-of-mouth multiplier for profitability). While these targets have reasonable validity, they capture only some of the drivers of film success. Other influential factors (marketing spend, production budget, and actor popularity) are deliberately excluded from the predictor set, narrowing the thesis to the specific question of physiological signal's relationship to perception and profitability.

Physiological reaction is also too broad of a concept to use for analysis. It is operationalised by only including signals collected by wearable devices. This is not only a restriction imposed on the thesis by data availability, but also a helping factor in terms of arguing for the implementation of the process at test screenings. In terms of scalability and availability, all these wearables prove to be optimal choices.

In terms of the connection between perception and physiology, this thesis does not aim to claim a causal relationship. What is tested is simply the predictive value that lies within physiological signals, and their association with movie profitability and reception.

In addition, feature selection from the raw data is hand-crafted. No training of deep learning models is performed on the original set of signals. Creating an architecture capable of learning from all the raw signals is out of the scope of this project. That is mainly excluded from the main scope because of efficiency and scalability concerns.

The introduction of feature extraction and model selection are in themselves narrowing the methodological scope of the thesis. The goal remains the same: capturing the signals in the data in a reproducible, scalable way with less resource consumption. Since the purpose of research is to see if this data is worth collecting and using for decisions in a test screening setting, raw video footage is excluded from analysis. Not only would that require much more resources such as computing power but would also be quite difficult to implement in practical terms.

Later on, when data simulation is introduced, the aim is to be able to present the pipeline in a setting with more data and with more complex models. It is a deliberate choice to produce synthetic data in a way that does not create new and false signals. It would also be possible to simulate more data, but that is deemed redundant as models already are expected to converge due to the low number of signals coming from the original data (because of the low number of detailed observations). Thus, analysis is expected to result in a data bottleneck which the

current thesis is not aiming to solve. What is sought after is learning as much as possible from the data available and testing pipeline/model scalability and performance.

Each movie is represented as a single vector of features. Analysis and modelling on time-sequence level and participant-level are excluded based on what fits the purpose of this particular research.

In general, lots of well-informed and theory-driven decisions are to be made in various areas, especially in data handling and analysis methodology. Those are reflected upon thoroughly and explicitly in the relevant upcoming chapters.

2. Literature review

There are various relevant research areas when it comes to the issue at hand. The implementation of a physiological data handling pipeline at test screenings requires knowledge not only about physiology, but psychology, business and machine learning as well. The complete coverage of such vast literature is outside the scope of research, the goal remains to find and report on the most relevant and well-established articles, summary studies and possibly identify a research gap that validates the thesis and its specific practical value. In the following, the most conceptually relevant papers are reviewed.

In a broad sense, physiology's correlation with emotion is a well-researched area. With technological advancement, it made possible to record, represent and process data of physiological signals, enabling further investigation of relevant dynamics (Picard, 1997). The work that followed was consolidated in a systematic review in 2010, establishing that distinct emotion classes produce different patterns of physiological signals (Kreibig, 2010). This finding provided a theoretical and established basis for further research in emotional recognition, a relevant area for the thesis as well.

Using film clips as stimuli for inducing emotions is not without research background either.

In 1995, such a dataset was validated (Gross, 1995), and in 2010 it was revalidated and expanded upon, further confirming film clip's capabilities as stimuli for research within the areas of physiology, emotions, psychology (Schaefer, 2010). The paper was also referenced by the research, basis of the dataset that this thesis uses (Saganowski S. K., 2022).

Another crucial development of technology, highly relevant especially for the business and practical part of this Master's thesis, was the transition from clinical, laboratory sensors to consumer-grade wearable devices for recording of physiological data. A systematic review of related research was published in 2022, identifying key challenges and bringing the topic up to date with advancements in machine learning and technology in general (Saganowski S. , 2022). The Emognition dataset itself was a type of benchmark supporting their research.

When it comes to the business aspect and consumer decision-making in connection to self-reports, a 2015 study provided a direct comparison between traditional self-report methods and physiological measures. Results indicated that the latter adds predictive value beyond what surveys and questionnaires could provide in themselves (Plassmann, 2015).

The combination of all these complicated research areas is what provides the theoretical foundation of this thesis. The single most relevant and closest paper that has enormous relevance to this project was released in 2015. They captured EEG data from participants

watching movie trailers and showed that these signals are usable for predicting not only individual preferences but also box office results for the full movies (Boksem, 2015). Another relevant paper from 2017 shows that EEG neural similarity (assessed on moviegoers watching trailers in movie theatres) proves to be useful in predicting content recall and box-office revenue of movies (Samuel B Barnett, 2017). This research itself was a continuation of a line of work started in 2012 which first established that physiological data from a small sample size can predict commercial success, in their case it was within the music industry (Gregory S. Berns, 2012).

In addition, extensive research supports the claim that the whole business purpose of this thesis lies on: that is, self-reports may not be enough in themselves to accurately capture the audience's reactions at test screenings. Already in 1977, a study demonstrated the issue of participants generating post hoc rationalisations that are plausible, in order to explain their thought processes (Nisbett, 1977). This general research was later on extended in a business-related, customer behaviour context, confirming that participants reasoning their product preferences had a negative effect on predictive validity of said preferences (Timothy D. Wilson, 1989).

In conclusion, the literature review shows extensive research in key areas relating to this thesis. This includes emotion recognition from physiological reactions, media success prediction and movie box office success prediction from physiological data directly. However, there is a distinguishable *research gap* that lies in the details which serves as the contribution of this master's thesis.

The explicit use case of physiological data recording, processing and analysis at test screenings is not a well-researched topic. The closest work to this thesis (Boksem, 2015) differs in modality, stimulus and scenario. Instead of EEG, this research uses a dataset based on wearable devices and builds a pipeline for processing and signal extraction. As opposed to trailers, actual film clips are used, more reliably capturing the style and cinematography of the actual film product, instead of trailers that are made specifically to be short-form marketing content. In addition, all of the directly relevant papers conducted their research in a lab setting, a completely different scenario in comparison to test screenings. This is important especially with regards to efficiency. Current literature is not focused on building an efficient pipeline and scalability, due to not focusing on direct and practical application. In a lab setting, it is completely feasible to process the amount of data collected. For a huge studio that is interested in reactions to whole movies, doing multiple test screenings across multiple audience groups, efficiency in data processing becomes a key factor. The thesis contributes by aiming to close the gap on both a methodological (RQ1: building the pipeline) and an empirical scale (RQ2: testing the pipeline by applying it on the Emognition dataset and interpreting results, while also augmenting the real data with additional simulated observations in order to tackle the challenge of low number of observations).

3. Data

This paper combines two vastly different data types and sources in order to examine the possibilities of using physiological reactions as valuable indicators of movie perception and quite possibly, movie profitability, although that is a much more multi-dimensional issue with lots of pitfalls that are to be explained. The physiological data is derived from the Emognition dataset, while scene-level and financial data is collected manually for the movies relevant in the aforementioned dataset. Putting it all together, and more importantly, finding an accurate and efficient representation of data is a crucial point of this research due to the extensive nature of the available data and the complexity of the issue at hand.

3.1 Emognition Wearable Dataset

The Emognition dataset (Saganowski, et al., 2022) was initially created in 2020 by researchers with the purpose of testing methods for emotion recognition from physiological responses and facial expressions. Participants were shown movie scenes while wearing 3 wearables with the purpose of recording physiological data.

Emognition dataset: Data Collection Devices

Device	Type	Position on the participant	Signals captured	Sampling rates	What it captures
Empatica E4	Wristband	Underside of wrist	BVP, EDA, IBI, ACC, TEMP	BVP 64 Hz, EDA and TEMP 4 Hz, ACC 32 Hz, IBI event-driven	Peripheral autonomic activity
Samsung Galaxy Watch 4	Smartwatch	Top of wrist	Heart rate, PP interval, BVP (raw and processed), ACC, gyroscope, rotation quaternion	HR and PPI 10 Hz, BVP 20 Hz, ACC and gyro 33 Hz	Cardiovascular activity, complementing Empatica
Muse S	EEG headband	Forehead and behind the ears	Raw EEG, pre-computed band powers (Delta, Theta, Alpha, Beta, Gamma) per channel, ACC, gyroscope, HeadBandOn quality flag	256 Hz for all	Central nervous system activity

Table 1 - Emognition wearable devices overview

Emognition dataset: Data Processing Software

Software	Type	Source	Signals captured	Sampling rates	What it captures
OpenFace 2.0	Toolkit applied to webcam video	Video recording of participant	Facial Action Unit (AU) intensities (17 AUs), head pose (3-axis rotation), gaze direction, tracker confidence	60 fps	Visible facial behaviour: smile dynamics, brows, lip-corner depress, blink, head movement, gaze stability
Quantum	Commercial emotion-classifier API on webcam video	Video recording of participant	Per-frame probabilities for six emotions (neutral, anger, disgust, happiness, sadness, surprise) and a dominant-emotion label, head pose	60 fps (per-frame)	Complementing OpenFace: emotion distribution dynamics and dominant-emotion transitions

Table 2 - Emognition data processing software overview

The stimuli and the extensive amount of data make this dataset ideal for this research, with certain limitations. There are also questionnaires that contain self-reports from which features are also derived. This may sound contradictory at first, but including self-reports serves the purpose of solidifying the claim of the thesis, which is the importance of physiological data. If analysis shows the overwhelming importance of self-reports, then the very purpose of this research can be questioned. Alternatively, if one of the key findings is that it is the physiological markers and reactions that matter, that may serve as a solid argument in favour of validity.

The Emognition dataset also contains full video recordings of the experiment. That means, for all 43 participants, 10 mp4 files are provided. This thesis does not utilize them, since the numeric data covers and records all signals. However, it is important to mention that a similar

problem could be framed as a computer-vision problem, certainly leaving room for further research. Regarding the participants, they are relatively young students. That unfortunately means the dataset cannot sufficiently represent the consumer population. This is an obstacle this current research cannot overcome, lowering the overall trustworthiness of the results and generalizability.

Methodologically, it is very important that a neutral washout clip is shown to the participants between every emotion clip, helping them return to baseline levels and avoiding leakage of data from their previously experienced scene.

The validity of the dataset may be questioned as the original purpose of data collection and research differs from what is proposed in this thesis. Originally, the aim was to recognize certain emotions, neglecting business aspects, hoping to understand physiology better. Thus, film clips were chosen based on previous research aiming to elicit different emotions. This is a key point which makes the dataset a reasonable choice to test the pipeline on: resulting stimuli vary in terms of style, music, and different cinematographic considerations. Later, manual data collection about the relevant movies also shows a wide representation of genres, budget levels, and resulting success metrics.

There are, however, two limitations that come with the dataset which somewhat lower the validity and narrow the possible scope of research. The Emognition research had 43 participants, all of which belonging to a specific demographic group. Arguably, that specific age group is of interest for film businesses, but the issue of the population not being correctly and thoroughly represented persists. In addition, the dataset does not contain movies that were complete commercial and critical failures. By using this dataset, the models are not able to learn about patterns and signals specifically contributing to why certain movies fail. What is more likely to be observed are the success factors and dynamics in relation to physiology, in movies from different time periods, genres, studios.

3.2 Scene and movie-level features

The rest of the non-simulated data used for analysis is collected manually. It is made possible by the fact that the Emognition dataset uses real movie scenes as stimuli, and the number of observations (given that we consider all participants' aggregated reaction as a single observation) is quite low. Therefore, adding new dimensions to already existing physiology-focused research generates new opportunities in numerous ways, especially in the areas of model training and business. The movies cover multiple genres and eras across the long history of motion pictures, which has its advantages and numerous biases and differences that must be addressed during the data collection and handling process.

The manually collected data can be divided into 2 segments: scene-level features and movie-level metadata.

The motivation behind collecting scene-level features lies within the very purpose of research. If the data and consequent results are to be used in a test screening setting, these are the factors that are relatively easy to change, since most apply to the post-production phase of filmmaking. Scene properties are recorded as categorical variables (low/moderate/high). This is a deliberate methodological decision, implemented with efficiency in mind. Arguably, one would get more accurate results and better scene differentiation if numeric, continuous values were used, but

in this research that option was discarded. That is because some of the properties, such as brightness and audio/motion intensity are very hard to quantify. In addition, the stimuli in the Emognition dataset vastly differ from each other, so even with this rather simple categorical representation, scenes were easily differentiated.

Each scene-level property was selected based on the premise that they may be a contributing factor in influencing the viewer's perception of motion pictures. They are key features that establish the essence and individually contribute to a clip. Film is an art form combining audio-visual and narrative storytelling; therefore, it is crucial to examine the choices in camera movement, music and dialogue. This part of the collected data serves solely as a complement to the physiological data and to shed light on possible connections. The full analysis and research of how these features each influence perception or profitability falls outside of the scope of the thesis.

Scene-level features

Feature name	Description	Encoding
cut_count	Frequency of cuts (shot changes) per clip	Categorical (low/moderate/high)
brightness	Average brightness level of the clip	Categorical (low/moderate/high)
motion_intensity	Average on-screen motion magnitude (camera and subject movement)	Categorical (low/moderate/high)
audio_loudness	Average audio loudness over the clip	Categorical (low/moderate/high)
silence_ratio	Fraction of clip duration containing silence or near-silence	Categorical (low/moderate/high)
music_presence	Extent to which music underscores the clip	Categorical (low/moderate/high)
dialogue_density	Dialogue density (proportion of clip with active speech)	Categorical (low/moderate/high)
face_screen_time_ratio	Proportion of clip in which a face (any character) is on-screen	Categorical (low/moderate/high)
lead_screen_time_ratio	Proportion of clip in which the lead character is on-screen	Categorical (low/moderate/high)

Table 3 - Scene-level features overview

Movie-level metadata consists of release year, genre, country of origin, IMDb (IMDb, 2026) rating, budget, and revenue data. In addition to that, a target variable - one besides IMDb rating - is calculated which is called WOM ("word- of-mouth") multiplier, which will be expanded upon in the relevant chapter of feature extraction and data manipulation.

Due to the complexity of the movie industry and the intertwined nature of art and business within the field, there are many biases to tackle and account for when dealing with this data in order to provide an accurate representation. These issues are most apparent when collecting financial data regarding movies.

First of all, budget data is more often than not unavailable for the general public. This is a complex topic with multiple reasons as to why budgets are kept a secret. The most important factors include upfront fees and profit participation agreements, drastic budget changes during the process, country-specific accounting practices and the very fact that the cost structure of movie production changes from era to era, making it rather difficult to compare (Follows, 2025). For this research, budget data unavailability is not actually a big issue. What this paper

investigates is the effect of physiological reactions on perception and profitability, not the ROI of a film itself. Therefore, due to the uncertainty and questionable relevance, budget data is only collected in the form of a categorical variable, similar to scene-level features. Even without knowing the exact numbers, objectively putting an indie, small-scale movie into a “low” budget category, and a Hollywood blockbuster into a “high” budget category is possible. It is acknowledged that this data does not carry as much significance as the exact numbers would, but it is deemed enough of a signal for this current research problem and business case. Accurate revenue data is rather hard to find, especially considering that the movies in question (stimuli used in the Emognition dataset) span over multiple decades, the oldest having been released more than 50 years ago. Setting up a relevant and well-suited outcome variable was a challenging task in itself. Simply considering all-time revenue contains lots of noise and uncertainty regarding the release era and marketing conditions. For this specific purpose researched in this thesis, influence of marketing and release era (streaming vs. non-streaming era) needed to be excluded as much as possible. After all, what the research question aims to shed light on is the effect of the movie products themselves, not the marketing campaigns that come before, or the revenue that comes from streaming revenue and re-releases that come after. As a result, after deliberating methodological consideration, only the worldwide gross revenue of the original release and the opening weekend revenue are considered for the calculation of one of the outcome variables. Revenue data is collected from Box Office Mojo, a website provided by the Internet Movie Database (Box Office Mojo by IMDbPro, 2026).

Only considering the original release helps dealing with the bias and effect of longevity. Had the aggregated revenue of all releases been considered, older movies would have considerably distorted values because of re-releases and newer ones as well, due to the effect of streaming service releases.

With the initial goal of the research in mind, it is crucial to also include opening weekend revenue as comparison. The first week of release is most likely to be influenced by factors not relevant for this research, such as marketing campaign and star power. Considering this and adjusting the outcome variable accordingly are meant to reduce the noise attributed to the aforementioned factors.

Movie-level metadata features

Feature name	Type	Description
movie_id	String identifier	ID used as a key identifier (M001-M010)
movie_title	String	Title of the film
imdb_link	URL	Direct link to the IMDb listing
targeted_emotion	Categorical	Emotion the clip was selected to elicit in the original Emognition research
clip_duration_s	Integer (seconds)	Length of the experimentally-shown clip
release_year	Integer	Theatrical release year
genre_primary	Categorical	Primary genre
genre_secondary	Categorical	Secondary genre
country_of_origin	Categorical	Country of principal production
budget_categorical	Categorical	Production-budget tier (low/moderate/high)

Table 4 - Movie-level features overview

It is important to mention that there were some anomalies during data collection, especially regarding one of the stimuli, “London 2012”. The nature of this clip is different due to the fact that it is from a miniseries documenting the events of the London 2012 Olympics. Therefore,

no financial data is available, and even if it was, it would be heavily influenced by the popularity and grandiose nature of the event it was based on. In the case of this non-theatrical documentary, a decision was made to impute these values instead of completely excluding it from the observations. The reason is that it is arguably worse to lose 10% of the real physiological data than to calculate the numbers instead in a way that influences the models the least later statistically. Thus, gross original release and opening revenue are imputed values calculated by taking the median value of the other 9 movies. This of course influences the later calculations of the actual outcome variable as well. There is a clear trade-off here which makes this solution imperfect, and admittedly, somewhat of a delimitation.

All the revenue-related figures need to be adjusted for inflation in general, since movie ticket prices and other various circumstances change depending on what release date a film has. Thus, there is a need for an index to account for some of this inconsistency and generalize the numbers. This solution has its own set of limitations, most importantly the fact that overall worldwide revenue is accumulated over long years, not just made at release year. Taking only the turnover of an original release serves to also partially account for this, but definitely not completely.

The chosen index to adjust revenue figures based on is CPI-U (Consumer Price Index for All Urban Consumers). There is a considerable number of methodological arguments as to why this choice was made: CPI-U numbers are government-issued and aggregate a wide variety of products and services, covering a good proportion of inflation. Most of the stimuli are also US releases (based on collected data of origin country). In addition, taking values from urban areas makes sense since most movie theatres are in places with higher population density. It also helps that even though the movies in question span over a large period of time (oldest being from 1978), CPI-U covers all of it. To be as relevant as possible during research, 2025 is used as base year (U.S. Bureau of Labor Statistics, 2026).

Original and inflation-adjusted theatrical revenue per film							
Movie ID	Title	Release year	IMDb	CPI-U (year)	Multiplier (322/CPI)	Original revenue (USD)	2025-adjusted revenue (USD)
M001	Three Colors: Blue	1993	7.8	144.5	2.228x	\$1,324,974	\$2,952,537
M002	Koyaanisqatsi	1983	8.2	99.6	3.233x	\$1,725,068	\$5,577,027
M003	Trainspotting	1996	8.1	156.9	2.052x	\$16,491,080	\$33,844,027
M004	Capricorn One	1978	6.8	65.2	4.939x	\$12,000,000	\$59,263,804
M005	American History X	1998	8.5	163.0	1.975x	\$23,875,127	\$47,164,361
M006	London 2012: Games of the XXX Olympiad	2012	7.4	229.6	1.403x	\$23,875,127	\$33,483,410
M007	Marie Antoinette	2006	6.5	201.6	1.598x	\$60,917,189	\$97,298,288
M008	The Blair Witch Project	1999	6.5	166.6	1.933x	\$248,639,099	\$480,562,964
M009	A Fish Called Wanda	1988	7.5	118.3	2.722x	\$62,493,712	\$170,101,228
M010	The Champ	1979	6.8	72.6	4.435x	\$30,441,738	\$135,017,075

Table 5 - Movie revenue calculation overview

3.3 Dependent variables

The choice of what target variables to use is crucial, as it directly impacts the results and overall conclusion of research. Since the research interests are revolving around perception and profitability, two separate variables are created. Causal relationships may not be established from the beginning; therefore, both are being predicted on even grounds.

First of all, it is important to establish why more standard metrics like all time revenue or ROI (return on investment) weren't chosen. Simply taking all time revenue (all releases) does not account for how long the movie has been on the market. This includes crucial differences in

distribution that happened over the years such as the rapid development of the DVD home entertainment market or streaming services that overtook the whole industry. In addition, as this research's main topic is physiological reactions to the movies themselves, the effect of marketing campaigns should logically be negated as much as possible. First-week sales revenue is mostly influenced by marketing, with the actual film product and its emotional impact having no effect whatsoever. This is a variable that needs to be adjusted for.

On another note, ROI values would be hard to compute in the context of this research. Issues persist not just on a conceptual level, but in practical terms as well. Data regarding production costs is mostly undisclosed, incorrect or simple speculation. Re-releases of certain movies on different platforms (DVD, Streaming) and in different qualities (4K) also cause problems of accurately assessing ROI and bring about uncertainty.

Having taken all these factors into consideration, two variables were chosen to represent two aspects of this greatly complicated topic. Firstly, as the primary dependent variable, IMDb rating was chosen. It is a continuous variable on a scale from 1 to 10. In reality, the rating distribution is rather skewed to the right (Movie ratings on IMDb, 2021), so the variance, or lack thereof, in ratings of the original Emognition stimuli (ranging from 6,5 to 8,5) does not cause that much of a concern. However, having no film with a significantly low rating is a definite disadvantage causing lots of signals in the population not to be covered. In theory, one could use other rating variables (RottenTomatoes, Letterboxd, Metacritic). IMDb was chosen because it is the most popular one, other rating sites being rather niche in comparison, therefore it arguably covers the overall population of movie-watchers better, not just enthusiasts. This outcome variable captures customer perception quite well, with data availability for movies from any age.

Secondly, to gain some insight into the immensely complicated topic of movie profitability, an additional outcome variable is computed. Simply using a vague numeric value such as revenue would introduce many problems and huge amounts of noise brought by different effects. Older movies are hardly comparable in terms of total revenue to a newer one released in the streaming era. A huge blockbuster seems very profitable compared to an independent release if we are only comparing revenue in relation to production costs. Total turnover is largely driven by marketing campaigns and star power of lead actors that bring customers to movie theatres regardless of how impactful or quality the actual film products are. Inflation is another issue making unadjusted values impossible to compare.

Therefore, it is crucial to critically consider as many revenue-related, social and psychological factors as possible and adjust the target accordingly. Having made certain methodological choices, a word-of-mouth multiplier (WOM) was computed. The idea stems from the very purpose of this research, to research the relationship between perception, success and profitability of films, excluding outside effects as much as possible. The variable is, in short, defined as a ratio of total revenue to opening weekend revenue. It is meant to exclude some of the effects of marketing by adjusting to opening week turnover that is highly influenced by irrelevant factors to this research. The actual values are rather tricky to collect and need some adjustments to reflect reality as much as possible. All numeric turnover values had been inflation-adjusted, but this is no perfect solution either. The values reflect accumulated revenue across the span of the film's lifetime on the market; therefore, the CPI-U index correction inflates the values compared to reality, giving an optimistic estimate. However, this is still

deemed to be better than leaving the values as they were, while also negating the advantages that new movies have in general: bigger market reach, population of consumers and overall popularity of the entertainment industry. Towards the total revenue, only original releases are counted. It is important to adjust for the longevity of certain movies on the market. Re-releases would inflate the revenue numbers and are mainly influenced by factors such as familiarity and marketing which the current research tries to avoid as much as possible. Thus, the WOM metric is meant to capture the profitability of films based on engagement and customer perception. The outcome is practically a ratio, but a log-transformed value is also calculated in order to make it more suitable for regression. The underlying social and psychological assumptions for constructing this variable are backed by existing research. Emotionally arousing events tend to stay more in our memory (Cahill L, 1998), and creative products eliciting positive emotions in customers tend to be more popular (Berger, 2013). The combination of these factors led to the methodological choice of introducing WOM as an additional outcome variable, in search of the connection between movie profitability and physiological reactions.

Outcome variables of perception and profitability

Variable	Type	How it is calculated	What it represents
imdb_rating	Continuous (regression target)	Public IMDb audience rating, collected from the IMDb listing for each film at the time of data compilation.	Audience perception of the full film, aggregated over the international IMDb rater base. Time-stable and marketing-independent.
wom_multiplier	Continuous (regression target)	Worldwide theatrical revenue + opening-weekend revenue (both inflation-adjusted to 2025 USD via CPI-U).	Word-of-mouth strength: how much the film over- or under-performs its opening weekend in the rest of its theatrical run. Cancels marketing-driven opening scale.
wom_multiplier_log	Continuous (regression target)	$\log_{10}(\text{wom_multiplier})$. Transform applied to symmetrise the distribution for regression.	Same construct as wom_multiplier but on a log scale, introduced for regression because the raw ratio is heavily right-skewed.

Table 6 - Dependent variables overview

4. Methodology

The following chapters contain the detailed explanation of how the thesis aims to answer the proposed research questions. Methodology is the direct answer to RQ1, the consequent implementation of it and testing it on actual models leads to RQ2 in the Results part.

Disclaimer about usage of Generative AI:

The thesis uses Generative AI for tasks related to Python analysis code framework creation, inspiration in relation to data simulation method, editable figure/table framework creation and documentation of used data. The official declaration document and the detailed explanation of usage cases and purposes can be found attached as an appendix.

4.1 General overview of the pipeline

The purpose of the pipeline is to handle and aggregate multiple data sources and find an efficient and accurate representation of the physiological data used for analysis. Part of the process is also combining it with additional metadata about scenes and movies. The figure below presents the whole process of getting from the Emognition raw dataset to model-ready data. Feature extraction is done on all five sources, plus self-reports. The original dataset works with 10 clips (of around 2 minutes) and 43 participants, which already amount to more than 40 gigabytes of data. Scalability is clearly an issue, and the pipeline provides a solution which can be feasibly applied to much more participants and much longer stimuli. It is a crucial step for the thesis to take because it directly helps to move the research from solely theoretical grounds towards practical implementation. It comes with business value that the pipeline would be able to handle physiological reactions to whole movies instead of just minute-long clips, with hundreds or thousands of participants, with the same collection method. This is a result of numerous methodological decisions and trade-offs that are to be explained in detail later on.

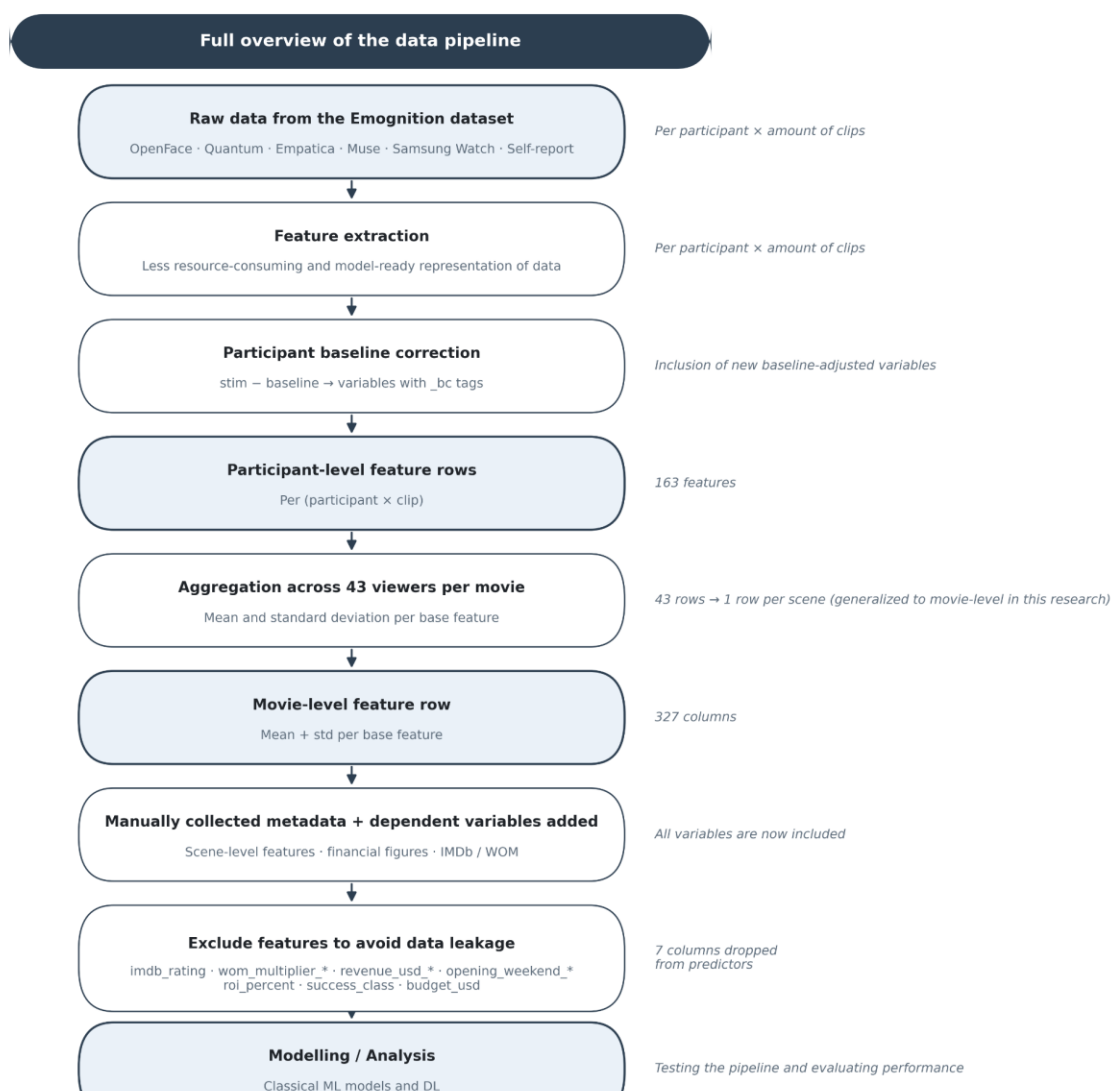


Figure 1 - Data pipeline overview

4.2 Emognition feature extraction and data manipulation

Feature extraction is a crucial step when dealing with such an extensive and complicated dataset like Emognition. Data collected on 43 participants over 10 movie scenes (on average 2 minutes per scene) already amounts to more than 40 GBs of data, excluding the video recordings. The scalability of such data collection methods is rather low, and the structure itself isn't necessarily optimal for ML model training due to varying data structures from different sources. Therefore, the need is established for a data processing pipeline. In the following, a detailed explanation is provided on the structure of the raw Emognition dataset, which features were selected from it, and how it was computed.

The raw Emognition dataset (excluding the video recordings) can be separated to four main categories: OpenFace (Facial Action Units and head poses), Quantum (data from an independent emotional classifier API), Study_data (Empatica, Muse and Samsung Watch data) and Questionnaires (self-report data after stimuli). This structure provides very good coverage of signals that may be relevant for modelling purposes but are also equally tremendous. Thus, it is crucial to have relevant and significant features selected from them and represented in a more sufficient format.

4.2.1 OpenFace features

During the initial experiment which the Emognition dataset is based on, participants watched film clips while a webcam recorded their face for the whole duration. As mentioned before, the raw video footage is fully available. In addition, during the dataset creation all the recordings had been post-processed with OpenFace 2.0, an open-source project to estimate Action Unit (AU) intensity, head pose and gaze direction at each frame during clip viewing (Baltrusaitis, 2019). Action Units indicate physiological reactions such as lip, brow movements, blink events. Each has their own AU code and interpretation which are part of the Facial Action Coding System (Ekman, 1978).

The raw file structure includes a CSV file for each session phase (a person watching one of the movie or washout clips) at 60 frames per second with more than 700 columns. Needless to say, this results in a huge dataset even in the case of a relatively low number of participants and movies. It is crucial to extract as much of the signal provided by these variables as possible while also drastically improving efficiency.

The chosen extracted features therefore cover all the main things that the original ones measure: smile dynamics, AU velocity, brow, blink dynamics, head movement and gaze stability. There is also an additional measure for confidence for each frame.

Features extracted from OpenFace (35 base at participant level × {mean, std} = 70 in total)

Group	Base features	What it captures	Source
Smile dynamics	12	Frequency, intensity, duration, onset latency, peak timing, and first- vs second-half buildup of smiling (AU12 lip-corner puller) and the Duchenne-smile subtype (AU12 + AU06 cheek raiser).	OpenFace AU12_r, AU06_r intensity time-series
AU velocities	4	Mean frame-to-frame change in selected AU intensity — rate of facial-expression dynamics for AU12, AU04, AU09, AU25.	OpenFace AU12_r, AU04_r, AU09_r, AU25_r
Negative-affect AU descriptors	12	For each of brow furrow (AU04), nose wrinkle (AU09), and lip-corner depressor (AU15): mean intensity, peak intensity, baseline-corrected mean, and onset latency.	OpenFace AU04_r, AU09_r, AU15_r
Blink dynamics	3	Blink rate per minute, baseline-corrected rate, and inter-blink-interval variability.	OpenFace AU45_r
Head movement	2	Mean head angular velocity (L1 norm of pose deltas) and fraction of clip with head velocity below a stillness threshold.	OpenFace pose_Rx, pose_Ry, pose_Rz
Gaze stability	1	Standard deviation of gaze magnitude across the clip (combined horizontal + vertical angle).	OpenFace gaze_angle_x, gaze_angle_y
Tracker confidence	1	Mean OpenFace per-frame confidence.	OpenFace confidence column

Table 7 - OpenFace features overview

4.2.2 Quantum features

This part of the dataset takes the same video footage as the basis. It is processed by a software called Quantum Sense with the goal of predicting and classifying elicited emotions. The raw output are probability distributions over six emotions and a dominant emotion label for each frame.

Extracted features include descriptive statistics for each emotion, engagement-related statistics, distribution metrics, and happiness-specific variables.

Features extracted from Quantum classifier output (29 base × {mean, std} = 58 in total)

Group	Base features	What it captures	Source
Per-emotion statistics	18	For each of six emotions (neutral, anger, disgust, happiness, sadness, surprise): mean probability, peak probability, and baseline-corrected mean = 6 emotions × 3 stats.	Quantum per-frame emotion probability vector
Engagement dynamics	2	Fraction of frames with non-neutral probability above 0.1 and time-to-first-non-neutral above 0.2 (engagement onset latency).	Quantum non-neutral probability sum
Distributional descriptors	3	Mean Shannon entropy of the per-frame emotion distribution, dominant-emotion transition rate per minute, and mean run length of the dominant emotion.	Quantum emotion distributions and dominantEmotion labels
Happiness-specific dynamics	3	Time-to-first-happiness above 0.1, linear slope of happiness across clip, and first-vs-second-half happiness buildup.	Quantum happiness probability time-series
Peak-emotion timing	1	Normalised position (0-1) of the frame with the highest non-neutral probability sum.	Quantum non-neutral probability sum
Head pose stability	2	Standard deviation of yaw and pitch across frames (independent measurement complementing OpenFa	Quantum headPose.yaw, headPose.pitch

Table 8 - Quantum classifier features overview

4.2.3 Study_Data features

The previous two sources explained in 3.3.1 and 3.3.2 are practically numeric representations and post-processing of the video footage recorded during the experiment. Study_data, on the other hand, captures entirely different physiological signals by collecting data from participants during the research wearing different devices. Based on these devices we may group the collected data into 3 categories: Empatica (from the Empatica 4 wristband), Muse (from the Muse S EEG headband) and Samsung (from the Samsung Galaxy Watch 4 worn during the stimuli experience).

Firstly, **Empatica** shows measurements in autonomic nervous system activity across 5 streams, each having their own sampling rate. Same as all the other wearable devices, monitoring is fully present during both movie clips and washout periods. Extracted features are also designed to represent signals from all 5 streams.

Features extracted from Empatica wristband (25 base × {mean, std} = 50 in total)

Group	Base features	What it captures	Source
EDA (electrodermal activity)	8	Mean tonic level, baseline-corrected mean, variability (std), linear slope, skin-conductance-response (SCR) rate per minute, baseline-corrected SCR rate, SCR onset latency, first- vs second-half buildup.	Empatica EDA stream (4 Hz, μ S)
HRV (heart-rate variability)	10	Mean HR, baseline-corrected HR, mean and std of inter-beat intervals (IBI), RMSSD, baseline-corrected RMSSD, pNN50, baseline-corrected pNN50, first- vs second-half HR reactivity, HR dynamic range.	Empatica IBI stream (event-driven, filtered 0.3–2.0 s)
BVP variability	1	Standard deviation of the raw blood-volume-pulse waveform.	Empatica BVP stream (64 Hz)
Skin temperature	2	Mean wrist temperature and its baseline-corrected version (proxy for peripheral vasoconstriction / sympathetic activation).	Empatica TEMP stream (4 Hz, $^{\circ}$ C)
Movement / fidgeting	4	Mean and baseline-corrected mean of ACC magnitude, std of ACC magnitude, and rate of ACC peaks per minute (fidget rate).	Empatica ACC stream (32 Hz, 3-axis → magnitude)

Table 9 - Empatica wristband features overview

The **Muse** headband has 4 electrodes, through which it collects all the data. Therefore, most of the variables in the raw dataset have 4 channels. Each session produces a JSON file with a 256Hz sampling rate. The electrodes are positioned at different places on the participant's head and are measuring different regions of their cortex.

Features extracted from Muse headband (20 base × {mean, std} = 40 in total)

Group	Base features	What it captures	Source
Frontal alpha asymmetry	5	Mean $\ln(AF8) - \ln(AF7)$, baseline-corrected mean, within-clip variability (std), linear slope, and first- vs second-half buildup. Indexes approach (positive) vs withdrawal (negative) motivation (Davidson 1992).	Muse Alpha_AF7, Alpha_AF8
Frontal beta asymmetry	2	Mean log-ratio of beta power at AF8 vs AF7 and its baseline-corrected change — auxiliary motivational marker.	Muse Beta_AF7, Beta_AF8
Frontal alpha & theta power	4	Mean frontal alpha power, alpha suppression (baseline — stimulus, positive = increased alertness), mean frontal theta power, and frontal theta change (emotional/cognitive-processing marker).	Muse Alpha_AF7/AF8, Theta_AF7/AF8
Per-channel alpha	8	Mean alpha power and baseline-corrected mean for each of four channels: AF7, AF8, TP9, TP10. Allows model to separately learn from each electrode beyond the AF7/AF8 asymmetry.	Muse Alpha_AF7, Alpha_AF8, Alpha_TP9, Alpha_TP10
Alpha variability	1	Standard deviation of frontal alpha across the clip — within-clip EEG dynamics without baseline reference.	Muse frontal alpha series

Table 10 - Muse headband features overview

Lastly, to complement Empatica’s wristband measurements, **Samsung** contains data collected from a smartwatch to measure cardiovascular activity. The actual, real-life positioning of this device differs from the wristband, leading to different signals. Similarly to before, one JSON is produced with 7 streams at varying sampling rates.

Features extracted from Samsung Galaxy smartwatch (15 base × {mean, std} = 30 in total)

Group	Base features	What it captures	Source
Heart rate	6	Mean BPM, baseline-corrected mean, standard deviation, dynamic range (max — min), first- vs second-half reactivity, and normalised peak-timing position.	Samsung heartRate stream (10 Hz, filtered 30–200 BPM)
PP-interval HRV	4	Mean PP interval, baseline-corrected mean, std of PP intervals, and RMSSD (Samsung-derived HRV).	Samsung PPIInterval stream (10 Hz, filtered 300–2000 ms)
Movement	5	Mean accelerometer magnitude, baseline-corrected mean, std of magnitude, accelerometer-burst rate per minute, and gyroscope-magnitude std (rotational movement unique to Samsung).	Samsung acc + gyr streams (~33 Hz, 3-axis → magnitude)

Table 11 - Samsung smartwatch features overview

4.2.4 Self-report questionnaires

After watching each movie scene, participants were asked to self-report and rate their emotions felt, providing SAM (Self-Assessment Manikin) ratings categorized into valence, arousal and motivation (Margaret M. Bradley, 1994). The final raw JSON structure also contains metadata on the participants that serve as additional context.

Even though the purpose of this research is precisely to go beyond self-reports and derive results and connections from physiological data, establishing it as a valuable source, self-reports cannot be neglected. In fact, it comes with value to directly evaluate their usefulness, which directly adds a degree of validity.

Extracted features include self-reports for all emotions that appear in the research, SAM dimensions data and context, in a more model-ready representation.

Features extracted from self-report questionnaires (21 base $\times$ {mean, std} = 42 in total)

Group	Base features	What it captures	Source
Discrete emotion ratings (raw)	9	Likert ratings (1-5) on each of nine emotions: amusement, anger, awe, disgust, enthusiasm, fear, liking, sadness, surprise.	Post-clip questionnaire (SR-EMO scale)
Discrete emotion ratings (baseline-corrected)	9	Same nine Likert ratings minus the participant's baseline-fixation ratings on the same emotions — clip-induced delta.	Post-clip questionnaire minus pre-clip baseline
SAM dimensional ratings	3	Self-Assessment Manikin (SAM) ratings (1-9) on valence (pleasantness), arousal (activation), and motivation (approach / withdrawal).	Post-clip SAM questionnaire

Table 12 - Self-report features overview

4.2.5 Aggregation from participant level to movie-level and baseline correction

In order to make the Emognition dataset usable for this thesis's research purpose, all features need to be aggregated to movie-level and combined with manually collected scene properties and revenue data.

It is one of the biggest known constraints of the paper that the stimuli are only 2-minute scenes from given movies, meaning all the data except for revenue only samples a very small portion of the films. This definitely questions generalizability and lowers the overall trustworthiness of the results. On the other hand, the scenes in question may still capture many features of the actual movie, including cinematography, music, stylistic properties and more. In addition, the relatively small length of the stimuli makes this research project feasible and somewhat scalable in the first place. This is another trade-off that needs to be considered and one that puts the complexity of this problem in perspective once again.

Firstly, all selected features are computed for all 43 participants over the 10 movie scenes. Then, for each numeric feature column the features are aggregated to movie level. In practical terms, this means each feature has a mean and standard deviation calculated. Standard deviation is added because they represent information regarding inter-participant agreement on the certain scenes. For example, a scene that polarised the participants, which may be considered controversial in nature, has lower mean and higher standard deviation for the arousal-related feature. This can help us understand more about the subjectivity aspect of different people watching work of art, carrying important, otherwise hardly represented signals.

Baseline correction is necessary to adequately represent the data. The Emognition dataset provides everything needed, as participants watched a baseline recording before each session (dots and lines on a black screen, completely lacking any kind of emotional content).

The final data structure used for modelling contains 320 features and 3 identifier columns. There are only 2 dependent variables (IMDb, WOM), but some categorical features are computed based on them to test some classifier models. The exact distribution of where the final features are derived from is as follows:

- 70 features from OpenFace (mean + std for each base feature)
- 58 features from Quantum (mean + std for each base feature)
- 50 features from Empatica (mean + std for each base feature)

- 40 features from Muse headband
- 30 features from Samsung smartwatch
- 42 features from self-reports
- 10 columns for data availability (for example has_OpenFace as a dummy variable)
- 20 variables regarding participant metadata providing extra context

4.3 Data simulation

After extracting the features and finishing the preparation of the final structure, a difficult methodological choice was made to introduce synthetic data, with the main goal of increasing the amount of data available for training.

The biggest reason why this step was a necessary addition is the number of observations was too little ($n=10$). In order to get valuable insights regarding the research problem, movie-level data is needed and the Emognition dataset provides only 10, in full detail. Models would not train effectively, train/test splits and cross-validation would practically be impossible. The pipeline provided in this paper provides a more efficient way of data representation, and generalization from huge amounts of physiological data. Though that certainly comes with business value, using only the Emognition dataset could hardly ever lead to meaningful results. Fortunately, simulating data based on the Emognition dataset has solid arguments for it. The initial stimuli were collected in a way that they invoked completely different emotions. This directly comes from their original research purpose of emotional recognition. This helps a lot because the movies in question cover a wide variety of genres, cinematic styles, topics and time periods. It gives a better overview and more varying signals regarding the movies and their emotional impact. The overall dataset, and thus the simulated additions both have the necessary variance to get meaningful results. On the other hand, variance in participants is admittedly lacking, but that issue stems from the raw dataset which this paper has no means to influence. Even though the basis for data simulation is given, proper execution is key. Sklar's theorem (Sklar, 1959) provides the framework for fitting the copula (Nelsen, 2006), a multivariate probability distribution from which sampling is done in order to produce synthetic feature vectors that match the original, real data's properties such as marginals and correlations (the latter just approximately). This method is a practical and established choice for tabular data with small number of observations (N. Patki, 2016). The simulation also follows the structure in a sense that synthetic participant data is created first per movie, then movie-level metadata follows, and lastly an aggregation happens. To illustrate the pipeline usage and to train machine learning models, firstly 40 movies are simulated. Later on, to test the application in a deep learning environment, 10.000 are created.

Simulating data enables the whole pipeline to properly work and helps to shed light on the value created in this research. Proper cross-validation and training/test splits are enabled, and chances for understanding underlying structures and signals are improved. On the other hand, disadvantages must be addressed as well. In reality, no new information is being created by the synthetic movies. Since the initial number of observations is low, there is a need for critique regarding the end results. Models may end up showing some results, and leading the research into certain directions, but the degree to which these are significant has to be constantly and explicitly assessed, critiqued.

4.4 Cross-validation (CV) and evaluation decisions

The fundamental issue of the low number of observations ($n=10$) is addressed in this chapter. This amount of real data is not enough for any of the standard CV practices. Therefore, the decision was made to apply a “two-track” strategy in the evaluation, making it possible to display pipeline capabilities while also accurately reporting on signals from the real data. In practical terms, this means that different practices are implemented for certain models based on what data they use and statistical feasibility.

One track (using only real data, $n=10$) uses Leave-One-Out CV, reporting on the empirical capabilities and signals contained in data obtained from the Emognition dataset. The other track which uses and evaluates on the combination of real and simulated data, uses k-fold CV in order to better represent pipeline capabilities. There are also hyperparameter tuning grids in addition (nested cross-validation), applied to the three strongest regression models based on previous analysis, using both IMDb and word-of-mouth outcome variables.

After standard model training there is also a Stacking Ensemble which combines them, with the goal of improving model and results quality. In that case, an 80/20 train-test split is performed. The method of training on simulated data and testing on the real observations is also tested in a classification and deep learning environment. In addition, analysis is done using random seeds, with results being saved in JSON files, providing for complete reproducibility.

4.5 Model selection and hyperparameter tuning

The following section contains a detailed explanation of what models were chosen to derive insights from, utilizing the model-ready data structure which is a direct result of the pipeline. The main goal was to test for the same signal with different models with different assumptions and biases, while also looking for insights and dynamics to answer RQ2. Since this is a modelling environment where the number of features is significantly higher than the number of observations, it is important to test different types of models and look for some sort of convergence of results as a form of validation.

4.5.1 Regression models

The table below gives an overview of the regression models used, the reasoning behind the choice, their purpose and tuned hyperparameters.

Regression models			
Model	Methodological reasoning	What it does	Hyperparameters tuned
Ridge regression	L2-regularised linear baseline, being able to handle correlated variables	Minimises squared error plus an L2 penalty on coefficients, shrinking all coefficients toward zero	Regularisation strength alpha with logarithmic grid search inside 5-fold CV
Elastic net	Works well in high-dimensional datasets, feature selection capabilities	Combines L1 (from Lasso) and L2 (from Ridge) regularisation in a linear model	alpha and l1_ratio with grid search inside 5-fold CV
Partial Least Squares (PLS) regression	Works best with high-dimensional regression with small samples such as Emognition	Projects features onto latent components, regresses in the low-dimensional latent space	n_components from 1 to 10 with grid search inside 5-fold CV
Random Forest	Non-parametric tree ensemble; captures non-linearities and feature interactions	Averages predictions from many decision trees, each trained on a bootstrap sample with a random subset of features per split	n_estimators, max_depth, max_features, min_samples_leaf with grid search inside 5-fold CV
Gradient Boosting (HistGradientBoostingRegressor)	Bias-reducing tree ensemble; complementary inductive bias to Random Forest (sequential vs parallel tree learning)	Fits trees sequentially, each correcting the residual error of the previous; histogram-based for computational speed	learning_rate, max_depth, max_iter, l2_regularization, min_samples_leaf with grid search inside 5-fold CV
Gaussian Process Regressor (ARD-RBF kernel)	Bayesian non-parametric regressor with calibrated uncertainty, the ARD kernel performs implicit feature-relevance selection	Models the regression function as a Gaussian process, ARD-RBF learns a separate length-scale per feature, downweighting irrelevant ones	Per-feature length scales, signal variance, noise variance optimised with marginal-likelihood maximisation

Table 13 - Regression models overview

In addition to the traditional regressors, two additional tests are performed. Firstly, the six models are combined into a single model using stacking, finding the optimal weighted combination of the regressions. The basis for introducing stacking is the very fact that the base models were selected so that the models have different biases and error terms. This is expected to result in useful combinations, improving results and being able to learn more complicated signals.

The other extension of analysis is to test and quantify each modality's (OpenFace, Quantum, Empatica, Muse, Samsung, Self-report) contribution regarding predictive power. Therefore, the 320 features are grouped, and the groups are removed in turn. Then a regression is evaluated for each combination of feature groups where one is excluded completely. That gives an idea about possibilities regarding the removal of certain feature groups.

4.5.2 Classification models

The following table displays all the classification models used during analysis, including the methodological reasoning, purpose and hyperparameters tuned.

Classification models

Model	Methodological reasoning	What it does	Hyperparameters tuned
Logistic Regression (Elastic net penalty)	Tests classification under simultaneous L1 + L2 regularisation	Maximises the log-likelihood of class labels under a linear model with combined L1 and L2 regularisation	C (regularisation strength) and l1_ratio with grid search inside CV
Logistic Regression (L2 penalty)	L2-regularised linear classifier, fits dataset properties and is often considered baseline	Maximises the log-likelihood of class labels under a linear model with L2 regularisation	C (regularisation strength) with grid search inside CV
Random Forest Classifier	Non-parametric tree-ensemble classifier; same inductive-bias rationale as Random Forest regression	Averages class probabilities from many decision trees trained on bootstrap samples with random feature subsets	n_estimators, max_depth, max_features, min_samples_leaf with grid search inside CV
Gradient Boosting Classifier	Bias-reducing tree-ensemble classifier; complementary to Random Forest with the same regression rationale	Fits trees sequentially, each tree correcting the previous prediction errors	learning_rate, max_depth, max_iter, l2_regularization, min_samples_leaf with grid search inside CV
PLS-DA (Partial Least Squares Discriminant Analysis)	Latent-variable classifier; the classification analogue of PLS regression, suited to $p \gg n$ problems	Fits PLS regression on dummy-coded class labels; classifies by thresholding the continuous prediction	n_components with grid search inside CV
Gaussian Process Classifier	Bayesian non-parametric classifier with calibrated probability outputs	Latent Gaussian-process function combined with a logistic likelihood to produce class probabilities	Kernel choice (RBF) and length-scale with marginal-likelihood maximisation

Table 14 - Classification models overview

Both target variables representing perception (IMDb) and profitability (WOMlog) are continuous by nature. However, to present the pipeline in a classification context and give a more interpretable business framing to what is being predicted, four binary targets are derived from a single composite success score that combines both channels equally, defined as $\text{composite} = 0,5 \times z(\text{imdb_rating}) + 0,5 \times z(\text{wom_multiplier_log})$ where z denotes the standardised score over the $n = 50$ augmented corpus. The composite captures both critical reception and word-of-mouth talkability in one dimension.

First, a simple median-split is done, and a top-quartile split with the goal of identifying top-rated and especially well-received movies. Then, two additional splits are computed with similar purposes. One differentiates top 20% revenue versus the rest, the other one aims to distinguish commercial failures (bottom 30% of revenue is categorized as such). The asymmetric split for identifying big successes and failures is a deliberate choice: arguably from a studio's perspective, it is more crucial to identify films that are likely to perform poorly.

In the case of classification models, there are extensions introduced to verify, strengthen results and robustness. This includes a permutation significance testing, where test labels are randomly permuted and evaluated, helping find a strong argument for the model classifying above chance. In addition, different outcome variable framings are introduced specifically tailored to the movie industry and the research topic. These smaller methodological additions serve to provide more thesis-specific justifications and more direct business implications. This helps answer the second research question and supports the overall goal of augmenting the process of test screenings.

4.5.3 Deep learning model

A deep learning architecture was implemented with the main goal of pipeline capability presentation. The expectation is that this will not result for significant additional gains over machine learning models due to data availability limitations. For this specific model, another 10000 movies were simulated (aggregated from participants same as before). Due to the nature of the data simulation method, no new

signals are created, putting the thorough and robust training of a neural network outside of the scope of this thesis.

Deep learning baseline model

Model	Methodological reasoning	What it does	Hyperparameters tuned
Multilayer Perceptron (4-layer feedforward)	Tabular deep-learning baseline. MLP provides a directly comparable input format to the classical models	Feedforward neural network with architecture 360 -> 256 -> 128 -> 64 -> 1. Each hidden layer applies Linear -> BatchNorm -> ReLU -> Dropout(0.3). Output layer is a single Linear unit (regression head)	Architecture chosen a priori (not grid-searched). Training hyperparameters fixed: AdamW optimiser (learning rate 1e-3, weight_decay 1e-4); cosine learning-rate schedule; dropout 0.3; batch size 128; 120-200 epochs with early stopping (patience 20-30); 10% of training data held out as validation for the early-stopping signal

Table 15 - Deep learning model overview

Given more data, neural networks would be the most applicable solution to such a high dimensional and complex problem. Given the limitations of this specific thesis, constructing a more complex architecture seems redundant. To cover a different set of inductive biases, it is important to explore possibilities within DL, but fine-tuning and model comparisons is outside the scope of this research.

5. Results

This chapter contains a detailed explanation of all the model and method results, and an overall synthesis of all recognizable patterns, dynamics and signals from the data. It serves as a direct answer to RQ2, the second research question that gives an insight into the business value side of the introduction of physiological data analysis in test screening practices. As a reproducibility measure, a random seed (=42) is set in the relevant Jupiter Notebook model scripts.

5.1 Simulated data validation

In the method chapter, the need for synthetic data was established as the initial number of observations was insufficient (n=10). The Emognition dataset is strong at capturing a wide variety of signals due to their data collection methodology (self-reports, 3 wearable devices, video recordings), but it is lacking in terms of actual amount of footage for this research purpose. Thus, it is a scalable option enabling the pipeline to be a useful addition but making the end results of analysis rather limited. This is another trade-off that the thesis acknowledges, but providing a direct solution is outside the scope of research. In order to verify that the implementation of the data simulation methodology was successful, 6 tests were introduced, most of which apply spot-checks on generated values.

Validation of data simulation

What is being tested	Test	Result
Marginal distributions match	mean and SD of 15 representative features across all 5 modalities	15 / 15 features pass
Correlation structure is the same	Frobenius similarity over 50 low-NaN features	similarity = 0.560
Range bounds are the same	80 features checked against real [min, max] with 0.1% tolerance	0 / 80 violations
Class separation is preserved	sign of (high-tier minus lower-tier) difference for 8 features	8 / 8 directions agree
No leakage	nearest-neighbour Euclidean distance	mean 12.95 vs real 17.97, min 9.61 vs real 12.27
Cross-table structure test	three synthetic CSVs (metadata, participant, movie-level) internally consistent	25/25 spot-checks pass

Table 16 - Validation tests for simulated data

The first test confirmed that the simulated features have the same distribution as the real ones, even with a strict pass criterion of being within 0,5 SD (standard deviation).

The second test aims to verify that pairwise correlations between physiological features are preserved in the simulated data via correlation matrixes (50x50 for both real and simulated data). By comparing the two, a similarity metric is computed which is 0,56 for the tested 50 features. In practical terms, this means the synthetic correlation matrix disagrees with the real original ones by 44% of magnitude. This is acceptable if we consider all the near-zero cells and those insignificant differences in an environment where 50x50 matrixes are compared.

In addition, range bounds are also checked for 80 features. This includes ratios (where the value must be between 0 and 1) and other numeric variables. This is a check that is expected to pass by definition: the data simulation method constrains the generated values to be within the range of sorted real values for all features.

It is also crucial to make sure that signals between success tiers are preserved. If a movie that has a higher IMDb rating has more smiling during reaction or increased hearth activity in comparison to a lower rated one in the real dataset, the same should apply for the simulated data. The test of the directions of relations ended as a success for all tested features.

The fifth test discards the possibility of leakage and duplications from the real data. Synthetic data comes from the same region of feature space, but it should not be direct copies of already existing observations. Minimum distance above 0 means there aren't exact duplicates, while the mean distance shows a comparable, reasonable difference. The results are positive, showing reasonable differences, not being too far from the real distribution while also avoiding identical values.

Lastly, some general level checks are implemented at once. All ID's across tables, number of participants and aggregations match. In conclusion, the validity and correctness of the simulated data is verified in a robust way.

5.2 Regression results

The table below provides a summary of all regression-related results. The row order is constructed in a way that first the more basic experimental configuration results are shown, followed by the tests and improvements built upon them. The columns contain ρ (rho), R-squared and Mean Absolute Error values for both outcome variables. P values aren't shown as the thesis aims to defend validity via convergence across models. Instead, Spearman ρ correlation coefficient measures to what extent the models get the ranking of films right, regardless of prediction accuracy.

Complete summary of regression model results

Configuration	Model	IMDb R ²	IMDb ρ	IMDb MAE	WOM R ²	WOM ρ	WOM MAE
Augmented, single-CV, tuned (n=50)	Random Forest	0.544	0.683	0.299	0.207	0.415	0.381
	Gradient Boosting	0.476	0.712	0.307	0.136	0.364	0.394
	PLS (n_comp=2)	0.472	0.68	0.309	0.19	0.385	0.402
	Ridge	0.472	0.668	0.317	0.147	0.358	0.406
	Gaussian Process	0.454	0.661	0.319	0.063	0.3	0.431
	Elastic Net	0.365	0.589	0.347	0.131	0.379	0.406
Real-only LOO, tuned (n=10)	Elastic Net	0.115	0.463	0.533	0.431	0.685	0.355
	Gradient Boosting	-0.225	-0.896	0.679	0.245	0.552	0.446
	Gaussian Process	-0.235	-1.0	0.678	-0.235	-1.0	0.573
	Random Forest	-0.252	-0.994	0.685	-0.244	-0.988	0.57
	Ridge	-0.26	-0.97	0.685	-0.254	-1.0	0.578
	PLS	-0.975	-0.713	0.898	-0.866	-0.782	0.711
Nested CV, leakage-free (n=50)	Random Forest	0.492	0.643	0.318	0.174	0.377	0.397
	PLS	0.427	0.649	0.329	0.19	0.385	0.402
	Gradient Boosting	0.369	0.609	0.341	0.079	0.261	0.417
Stacking, single 80/20 fold — train	5 base + Ridge learner	0.873	0.983	0.164	0.719	0.981	0.244
Stacking, single 80/20 fold — TEST	5 base + Ridge learner	0.631	0.742	0.262	0.018	0.079	0.438
Ablation Config 1 — physiology only (258 features)	Random Forest	0.479	0.706	—	0.181	0.401	—
	PLS-tuned	0.485	0.703	—	0.184	0.4	—
Ablation Config 4 — full ablation set (340 features)	Random Forest	0.543	0.677	—	0.201	0.424	—
	PLS-tuned	0.463	0.668	—	0.189	0.379	—

Table 17 - Regression results summary

The first 12 rows show the results of the 6 initially selected base models, each in 2 different setups. Augmented tests include simulated data, and 5-fold cross-validation. The others use real data only with leave-one-out CV, in order to test how much there is to be learned solely from the Emognition dataset and possibly compare it to the augmented versions. The most crucial result here is the fact that the R² of IMDb converges within 5 of 6 models into the range of 0,45 and 0,55. In such a high dimensional setting, using models that have different biases and failure modes, this is considered to be strong evidence.

The PLS model, across all setups (that are re-tuned each time separately) finds the signal at 2 components. This indicates a low-dimensional relationship between physiology and IMDb rating, with few influential factors and many correlations and noise.

WOM (word-of-mouth multiplier) results are weak across all basic setups, concluding that this outcome variable may not be predictable based on physiologic data. In general, models trained and tested on real data only failed, verifying the need for data simulation and further augmentation.

In rows 13-15, nested CV is implemented for the 3 best-performing machine learning models. Nested cross-validation helps avoid leakage by tuning the hyperparameters inside each fold of CV, using the training data of that fold only, leading to unbiased test results. This is an improvement in comparison to the optimistic experimental models, increasing validity and generalisation to new data. RF performs best at IMDb R² = 0,492, Spearman ρ = 0,643. Removing hyperparameter leakage caused about 0,05 drop in R² values across the models.

Rows 16 and 17 contain the results of model stacking. IMDb test R² = 0.631 is a great result that is substantially limited by the single-fold draw it was evaluated on. In terms of validity, it certainly ranks lower in comparison to the nested CV version. When it comes to WOM, the train-test gap of R² is enormous. This shows that the base models achieve a good training fit, but completely fail to generalize to unseen data. The same gap is not as big in case of IMDb, which is a valuable insight in itself.

Stacking meta-learner coefficients

Base learner	IMDb meta-weight	WOM meta-weight
Random Forest	+0.777	+0.365
Ridge	+0.027	+0.547
PLS	+0.079	-0.030
Elastic Net	+0.055	-0.103
Gradient Boosting	-0.088	-0.185
Intercept	+1.142	+0.690

Table 18 - Model stacking results summary

Upon close inspection of the meta-learner coefficients, it may also be concluded that Random Forest dominates model stacking, other base learner coefficients being near-zero.

Rows 18-21 contain results from the two endpoint configurations of the ablation study. It is important to emphasize the reason behind the choice of selecting RF and PLS. Not only do these models have top performance in previous regressions, but they completely differ in failure modes and biases. RF is non-linear, prone to overfitting feature interactions in smaller datasets. PLS is fully linear, with the possibility and risk of missing non-linear signals completely. By running the ablation study with these 2, the aim is to reach convergence, similar results and increase validity that way.

Summary of all ablation configurations

Configuration	Features	RF IMDb R ²	PLS IMDb R ² (tuned)	PLS IMDb n_co	RF WOM R ²	PLS WOM R ² (tuned)	PLS WOM n_comp
1. Physiology only	258	0.479	0.485	2	0.181	0.184	1
2. + Scene	267	0.486	0.479	2	0.188	0.182	1
3. + Movie metadata	298	0.475	0.544	2	0.211	0.098	10
4. + Self-report (full)	340	0.543	0.463	2	0.201	0.189	1

Table 19 - Ablation test results summary

There are patterns that we may observe based on the results. Firstly, scene metadata does not contribute much, there does not seem to be many new signals that aren't already captured by physiological features. The addition of movie metadata improves PLS results, but does not contribute substantially in the case of RF. This finding indicates that PLS was able to capture linear signals from these features that the random forest already learned implicitly from physiological data. Adding self-reports does cause an improvement for RF but makes PLS results worse, leading to the conclusion that while physiology proves to be a universally useful feature class, self-reports are model-dependent. The most significant result in terms of practical implications, serving as a positive indication towards the original research purpose, is that physiological features alone were capable of extracting most of the signals from this dataset, preserving the full prediction ceiling on average within these models.

5.3 Classification results

Similarly to regression models, a summary table is created to display classification results and tests. Besides the evaluation metrics, model setups are grouped based on class definition and contain a row explaining what data the model is being trained and tested on.

Overview of all classification models and tests

Evaluation	Setup	Model	Accuracy	F1	Precision	Recall	ROC-AUC
Median-split	Augmented 80/20	LogReg-ElasticNet	0.9	0.889	—	—	0.88
	Augmented 80/20	LogReg-L2	0.9	0.889	—	—	0.88
	Augmented 80/20	PLS-DA	0.9	0.889	—	—	0.88
	Augmented 80/20	GP Classifier	0.9	0.889	—	—	0.84
	Augmented 80/20	Random Forest	0.9	0.889	—	—	0.8
	Augmented 80/20	Gradient Boosting	0.8	0.8	—	—	0.76
Top-25% hit	Augmented 80/20	PLS-DA	0.9	0.857	0.75	1.0	0.905
	Augmented 80/20	LogReg-ElasticNet	0.8	0.75	0.6	1.0	1.0
	Augmented 80/20	LogReg-L2	0.7	0.667	0.5	1.0	0.952
	Augmented 80/20	GP Classifier	0.7	0.0	0.0	0.0	0.952
	Augmented 80/20	Random Forest	0.7	0.0	0.0	0.0	0.857
	Augmented 80/20	Gradient Boosting	0.6	0.5	0.4	0.667	0.81
Real-only LOO	Real data only	LogReg-ElasticNet	0.6	0.714	0.556	1.0	0.68
	Real data only	GP Classifier	0.4	0.0	0.0	0.0	0.5
	Real data only	LogReg-L2	0.3	0.222	0.25	0.2	0.24
	Real data only	PLS-DA	0.3	0.222	0.25	0.2	0.24
	Real data only	Random Forest	0.3	0.222	0.25	0.2	0.16
	Real data only	Gradient Boosting	0.5	0.0	0.0	0.0	0.1
Synth→Real median transfer	Train on simulated, test on real	PLS-DA	0.8	0.8	1.0	0.667	0.729
	Train on simulated, test on real	Gradient Boosting	0.6	0.5	1.0	0.333	0.875
	Train on simulated, test on real	LogReg-L2	0.7	0.727	0.8	0.667	0.708
	Train on simulated, test on real	GP Classifier	0.6	0.667	0.667	0.667	0.708
	Train on simulated, test on real	LogReg-ElasticNet	0.6	0.6	0.75	0.5	0.625
	Train on simulated, test on real	Random Forest	0.6	0.5	1.0	0.333	0.542

Table 20 - Classification results summary

Firstly, experimental classification modelling results using median split (above split are considered successful below split are not) look promising, but are also insignificant in themselves. In case of Top-25% prediction, PLS-DA gives the most balanced results, while logistic regression seems to overpredict positives (at a precision of 0,6) and random forest and Gaussian process classifier both predict 0 positives. This Top25% evaluation setup differs from the business-related outcome variable framing of hit detection. The purpose here is only to show that median-split classification used for reporting on the most significant results generalise across thresholds. This is confirmed as models do not collapse even if the cutoff threshold for success criteria and binary split is changed.

Using only real data from the Emognition dataset fails statistically, which is expected behaviour considering the low sample size. This result further establishes the need for more data and the necessity of data simulation in this thesis. When implementing the “train on simulated data, test on real” data

configuration, PLS-DA prevails again due to its balanced results. Random forest is the only failure with an AUC of 0,542.

The permutation test constructed an empirical null distribution by shuffling the composite-median class labels 200 times and refitting the LogReg-ElasticNet (due to its previous performance and strong L1 regularisation making it harder to overfit the random labels) pipeline on each shuffle while keeping hyperparameters fixed at the real-data tuned values ($C = 0,023$, $l1_ratio = 0,1$). The original classifier achieved 0,900 accuracy: 4 of 200 random label permutations reached the same threshold, resulting in $p = 0,020$. This serves as a significant claim and increase of validity for the 90% accuracy reported previously.

The additional business framings of outcome variables (hit detection, flop avoidance at bottom 30%) that aimed to provide more interpretable results were also evaluated on all classification models, using a configuration of training on simulated data and testing on real observations.

Business-specific outcome framing results

Task	Model	Accuracy	F1	Precision	Recall	ROC-AUC	PR-AUC
Simulated to real transfer — hit (train 40 synth, test 10 real)	Gradient Boosting	0.8	0.5	1.0	0.333	1.0	1.0
	Random Forest	0.7	0.0	0.0	0.0	0.81	0.589
	LogReg-L2	0.6	0.0	0.0	0.0	0.762	0.556
	GP Classifier	0.6	0.0	0.0	0.0	0.762	0.556
	LogReg-ElasticNet	0.8	0.667	0.667	0.667	0.667	0.514
	PLS-DA	0.6	0.0	0.0	0.0	0.619	0.421
Simulated to real transfer — flop (train 40 synth, test 10 real)	GP Classifier	0.6	0.333	0.333	0.333	0.762	0.533
	LogReg-ElasticNet	0.6	0.333	0.333	0.333	0.714	0.633
	Random Forest	0.6	0.333	0.333	0.333	0.714	0.5
	LogReg-L2	0.6	0.333	0.333	0.333	0.667	0.467
	PLS-DA	0.6	0.333	0.333	0.333	0.667	0.411
	Gradient Boosting	0.7	0.667	0.5	1.0	0.571	0.383

Table 21 - Business-framed classification results summary

Most importantly, models trained on simulated data generalize well to real test data, predicting above chance for both tasks, which contributes to the justification of data simulation. AUC ranges from 0,619 to 1,0 in case of Top 20% hit detection, with 4 classifiers not predicting positive results, defaulting to assigning movies to be non-hits. LogReg-ElasticNet provides rather balanced results, but with quite average AUC. The flop detection AUC ranges from 0,571 to 0,762. 5 of the classifiers completely commit to positive predictions resulting in the same metrics. The critical insight here is that multiple different model architectures were able to find a common decision boundary for flops, while the same did not occur in the case of hit detection.

5.4 Deep learning results

The multilayer perceptron deep learning architecture is evaluated for both numeric outcome variables in 2 setups: 5-fold CV with the additionally simulated 10000 synthetic movies, and synth to real transfer which means training on simulated data and evaluating on real data.

MLP results

Evaluation	Target	R ²	Spearman ρ	Pearson r	MAE	RMSE
5-fold CV (n=10,010)	IMDb	0.667	0.752	0.861	0.282	0.344
5-fold CV (n=10,010)	WOM	0.239	0.476	0.516	0.451	0.574
Synth→real transfer	IMDb	0.761	0.823	0.873	0.293	0.34
Synth→real transfer	WOM	0.331	0.552	0.613	0.452	0.494

Table 22 - Deep learning model results summary

Similarly to standard machine learning models, MLP results also indicate IMDb rating to be the stronger target with R^2 values being 0,667 and 0,761. Evaluation on the real Emognition dataset ends up resulting in higher R^2 which is not the expected outcome.

WOM remains the target variable whose variance is a lot harder to capture from physiological data, with R^2 values of 0,239 and 0,331 across the two setups. The asymmetry of results seen in regression models remains. The inclusion of a deep learning baseline model confirms the data bottleneck influencing the entirety of analysis. The introduction of 10000 synthetic samples (meaning a 200x scaling from the original 50) for training only resulted in an R^2 increase of approximately 0,13. The prediction ceiling is reached across different model architectures and feature-engineering setups and is considerably set by the amount of movie clips in the original Emognition dataset ($n=10$).

5.5 Cross-experimental synthesis of the results

The pipeline aimed at making physiological data handling less resource-consuming and providing an accurate representation of multimodal data has been tested in various scenarios: standard regression models, classifiers and deep learning architectures. It is important to provide a thorough summary in order to provide an extensive and well-informed answer to RQ2, investigating whether there are any insights to be gained from the Emognition dataset in the areas of audience perception and movie profitability. There are several tendencies and cross-model dynamics observed from the results, and they are summarized in this chapter.

Physiology and audience perception (using the target variable of IMDb rating) has a strong, method-flexible predictive relationship. PLS at $n=50$ reaches $R^2 = 0,63$, GB performs similarly at 0,65. The MLP deep learning baseline at $n=10.010$ achieves 0,76. Across all models and methods used for testing, IMDb R^2 is within 0,63-0,76. With the implementation of an improved, nested cross-validation practice, the same still holds true and the actual change of R^2 value remains below 0,05. In addition, it is observed that Spearman ρ exceeds Pearson r , which exceeds R^2 in every regression-related setup. The pipeline succeeds at ordering the films better than recovering the actual ratings.

The relationship between physiology and profitability (represented and made usable by the introduction of the WOM variable) is weak consistently across different setups. R^2 values range between 0,2-0,33. The approximately 200x training dataset increase introduced for the deep learning model also only resulted in an improvement of 0,08.

The IMDb R^2 values are approximately 2 times higher than WOM R^2 values across all models, data setups. Thus, this asymmetry is not a property of a specific method, but preserved throughout the analysis part of the thesis.

Regarding the business framing of classification (hit movie detection and flop detection), results indicate that commercial failure is more detectable than success. Five out of six classifiers reach AOC=0,81 on the flop identification, while AOC is 0,75 for hit detection with four out of six classifiers predicting no positives. This difference persists across thresholds and validation regimes such as real data only leave-one-out CV, or train on simulated data and test on real data evaluation.

In addition, signals to the models come mainly from physiological data. The results of quantifying the relative contributions of certain data types show that physiology alone preserves 88% of random forest model performance and 105% of model performance of a tuned PLS model relative to the complete feature set. The combined contribution of the remaining groups of scene, movie metadata and self-reports is 12% (RF) or 0% (PLS). The subjective self-reports submitted by participants after viewing the movie scenes add only incremental value and signal within the boundaries of the Emognition dataset.

Lastly, a crucial and cross-model finding of the pipeline application is the prediction ceiling that is achieved by different setups. Five different classifiers reach the exact AUC of 0,81. The optimized PLS reproduces $n_components = 2$ under different augmentation and partitioning. By the introduction of a deep learning model and a vastly increased training set, IMDb and WOM R^2 only increased by 0,13 and 0,08 respectively. This serves as a confirmation that the thesis faces a data bottleneck, rather than model capability issues.

In conclusion, to directly answer RQ2, some key findings derived from applying the pipeline to the Emognition dataset include:

- Physiological data predicts IMDb scores much better than profitability.
- At the strongest model setup, the pipeline succeeds at IMDb ordering at $\rho = 0.82$ from simulated-only training. In every regression configuration the pipeline is better at ranking films than estimating actual numeric ratings.
- The pipeline is able to classify commercial failures much more reliably in comparison to huge successes.
- Predictive signal from the complete set of features is concentrated mainly in the physiological channels.
- The pipeline reaches the prediction ceiling from different models with different biases, due to the low number of real-data observations ($n=10$) resulting in a data bottleneck.

6. Discussion

This section of the thesis aims to discuss the results, patterns identified previously in detail, assessing the answers to RQ1 on criteria such as usability, validity, and relevance. In addition, the contribution of the thesis is explained, critiqued, and compared against existing literature. The cross-experimental synthesis chapter established 5 patterns that were found consistently across different types of models in regression, classification and deep learning.

Firstly, it is a clear observation from the analysis that perception (IMDb) is predicted much (approximately 2 times) better in comparison to WOM (profitability). This tendency continued across models and changes in training-set sizes. There can be both conceptual and methodological reasons for this. Firstly, the choice and construction of the variables highly impact the results. IMDb ratings are relatively closer in time to the viewing experience (the very thing physiological data records), while WOM has a much longer time distance, which means it includes different biases and trends, unknown variables that are impossible to account for in this research. The way WOM is calculated was supposed to discard some of these (problems regarding re-releases, different eras of release, marketing campaign influence), but results indicate that it still is rather hard to predict. The thesis-wide issue of data availability also limits the results: while the film clips used as stimuli cover a wide variety of genres, styles and release time periods, none of them were very unsuccessful critically and commercially. The whole distribution is not accurately represented, which limits the validity of the actual practical results and implications. The generalization of film clips to whole movies is also questioning validity: while it makes the pipeline efficient and the clips themselves arguably represent the respective movies well, recording data from whole movie viewing sessions would result in more signals, better and more significant results. It may also be a contributing reason as to why WOM results are worse: WOM represents and captures recommendation behavior and overall impact of a movie, making it much harder to capture from just few-minute clips. Despite the serious limitations in numeric results, the pattern itself is consistent with previous literature to an extent. In a broader sense, commercial success is shown harder to be predicted than perception (Boksem, 2015). Relevant papers used individual movie trailer preferences and box-office returns and came to the same conclusion. A 2017 paper (Samuel B Barnett, 2017) relied on content recall against revenue and reported revenue prediction to be harder. Though the constructed outcome variables differ, the pattern remains in case of different data sources as well. For the specific scenario of test screenings in this thesis, the conclusion regarding this result is that perception prediction should be the primary goal and profitability is secondary. Both carry business value but the extent to which they can be interpreted solely based on physiological data and the introduction of this pipeline must be explicit.

Another consistent observation across the regression models is that the pipeline recovers the relative ordering of films better than the actual IMDb ratings. The small number of real observations is the biggest conceptual contributing factor, but the result comes with serious practical implications relevant to the research purposes of this thesis. For the purpose of augmenting movie test screenings with physiological data with the purpose of assessing future perception and profitability, ranking may prove to be exceptionally useful. Movie studios can gain crucial business information upon comparing physiological data from different movies in earlier stages of distribution. Being able to identify which project will do better on the market,

even if the exact degree of success is not reliably quantified, can lead to well-informed decisions regarding resource-allocation and development priorities. Which version of a cut has better performance, which concept does better at test screenings, which audience target segment responds better to a movie. All of these decisions require relative ordering, which the pipeline performs well. The strongest configuration within the analysis is, $\rho = 0,82$, $r = 0,87$, $R^2 = 0,76$ for the MLP on simulated data training and real data testing for IMDb. Both ρ and R^2 values indicate practical usability value for studios at test screenings.

Results across classification models identify a pattern in which commercial success prediction proves to be harder than flop detection. This pattern exists not only through different models, but different thresholds and evaluation setups. Conceptually, this may be underlined by the dynamics of engagement and different physiological reactions produced across varying emotion categories. A film or film clip that fails to elicit emotions, resulting in low arousal and engagement can more easily be identified and categorized to have lower commercial success. However, to identify a film with high likelihood of commercial success using physiological reactions, there are many more signals to capture. Different genres achieve higher arousal by different means. A viewing experience of a slow-burn drama highly differs from a horror movie, and in order to understand the underlying dynamics, one would have to treat the whole experience as a time-series observation. Due to the nature of this thesis, using clips and rather low number of observations, the full understanding of such patterns remains outside the scope of the project. However, this limited result also has practical implications for test screenings. The data and the results indicate that the pipeline is able to reliably extract signals contributing towards a movie's unsuccessfulness. This is crucial business information for studios in terms of risk-assessment and management. For example, from a purely business perspective, being able to realize a movie is not likely to succeed before the distribution procedures and agreements can lead to decisions that may save a lot of money and cut losses for the producer. Even though the need for experimenting with the addition of physiological data beyond simple self-report practices at test screenings had already been established, an ablation experiment was conducted on the data available for this thesis in order to verify it. The relative contribution of each feature type was quantified using the best-performing models. Results support previous literature on the limitations of self-reports, emphasizing the importance of physiological reaction variables which alone preserve 88% of the performance in the case of the random forest model. This serves as a solid argument in favor of the research purposes of this thesis: while self-reports contain important signals, augmenting the test screening process with physiological data increases predictive value. It is also crucial to establish the limitation of the Emognition dataset based on which this conclusion was drawn. In the dataset available, post-clip ratings were given on emotional dimensions, which is not equal to more advanced interviews, surveys and written feedback methods that film studios use. Current film industry practices may capture more signals in relation to perception and profitability prediction, so in a real-life application the gap in relative contribution between physiological features and self-reports would most likely be smaller.

In addition, there are certain signals and parts of data within the Emognition dataset that are not being used in this current research. The seasonality component which may contain crucial tendencies is not fully represented in this pipeline, which becomes a more significant issue if the process is applied to physiological reactions data of full-length movies. The actual raw

video footage of participants viewing the clips is also available, without being processed by OpenFace or Quantum Sensor. Therefore, the whole research may be framed as a computer vision problem with dedicated deep learning infrastructure development. Mainly due to infeasibility and lack of data this option is outside the scope of research.

In the analysis code, some models assign feature importance, which isn't explicitly discussed in this paper. The reason for that is the issue of result validity and the fact that the purpose of this research is to facilitate the inclusion of a physiological data pipeline to test screening processes.

Lastly, the thesis-wide limitation and bottleneck of data is addressed. The $n = 10$ constraint affects all results and claims and means that the patterns observed only apply to the properties of this specific Emognition dataset. Generalizability remains an issue that the thesis does not aim to solve. All results and interpretations must be interpreted under these limitations. The same applies to using movie clips instead of full-length movies as stimuli. Testing the generalizability of signals captured from short clips to whole movies is outside the scope of this research. What the thesis argues for is a trade-off between efficiency and result quality, but it is undeniable that physiological data gathered from complete watching experiences of participants would lead to better and more valid results. Despite all the binding constraints of data and the bottleneck it meant during the modelling processes, the Emognition dataset has serious advantages. Firstly, by providing both video recordings and physiological data from multiple devices through many channels, their method of collection provides a wide variety of ways to capture signals in the data. Furthermore, the fact that their whole research can easily be implemented in a test screening scenario (due to the use of consumer-grade wearable devices), they perfectly fit and enable the business case of this thesis. For film production studios to even think about implementing such an augmentation to their test screening processes, it must be deployable and scalable using publicly available technology. This makes the Emognition dataset a suitable choice, which is further supported by this thesis by creating an efficient pipeline and a framework for modeling and interpreting results.

7. Conclusion

This thesis researched the possibilities of augmenting test screening procedures, complementing self-reports with physiological reaction measurement, processed by a pipeline to interpret results and gain relevant business information for decision making. In order to provide an informed answer, the problem is decomposed into two research questions. RQ1 addressed the pipeline design itself, while RQ2 focuses on the empirical insights gained by the practical usage of the developed pipeline on the Emognition dataset. This chapter focuses on the direct answers the thesis provides to these research questions and to the problem at hand in general, while proposing areas of further research and improvements.

RQ1 asked how a scalable and theory-driven pipeline can be designed with the goal of extracting and evaluating predictive signals of audience response to film clips in form of physiological data. The methodology chapter which contains the full pipeline and its building procedure contains the direct answer. Firstly, multimodal feature extraction is done from various sources in connection to wearable devices, video footage, self-reports and scene/movie-

level metadata. The resulting final table contains 320 independent variables (70 OpenFace, 58 Quantum, 50 Empatica, 40 Muse, 30 Samsung, 42 self-report, 10 availability dummy variables and 20 participant metadata context variables) computed at movie-level by aggregating participant signals using means and standard deviations. Then, in order to address the issue of low number of observations, synthetic data is created using a copula-based approach. This expands the available data from 10 real movies by creating data for 40 simulated movies in case of regression and classification models and 10 000 simulated movies for the deep learning baseline model. The goal of data simulation is to preserve marginal distributions and structures while enabling different cross-validation methods that would otherwise be impossible to implement at $n=10$. The pipeline then proposes a theory-driven selection of models for various purposes and with various biases for both machine learning and deep learning. The validation design uses two different tracks, one applies leave-one-out cross-validation on the 10 real films only, while the other includes the simulated data ($n=50$) and uses k-fold cross-validation. In addition, nested cross-validation, permutation testing for significance and an ablation experiment is implemented for different and more problem-specific research purposes. All these design decisions and components produce converging results, while supporting and presenting scalability using a deep learning architecture. Proposed efficiency is also made possible: the pipeline significantly reduces the size of the dataset while aiming to preserve the signals and providing a suitable representation of data for modelling purposes and business-ready interpretations.

RQ2 asked what insights into the relationship between physiological reactions to film clips and the corresponding films' audience perception and profitability can be gained by applying this pipeline to the Emognition dataset. The general patterns identified as results of the analysis provide an answer to this part of the question, which strengthens the argument for practical application of this pipeline and a real-life business context. Physiology is shown to predict audience perceptions much more reliably in comparison to profitability. IMDb prediction performance converges around $R^2 = 0,60 - 0,76$ across different model architectures, while WOM stands at around $R^2 = 0,2 - 0,3$. This remains the same with training set size changes and is probably bound by data availability and outcome variable construction limitations. The pipeline also consistently overperforms at relative ordering of IMDb ratings at $\rho = 0,82$. This identifies another valuable real-life use case in relation to test screenings, helping decision-making processes that require accurate ordering and comparison as opposed to direct and absolute value predictions. In addition, the pipeline's most significant classification capability lies in detecting commercial failures. This is another useful use case for practical purposes for film studios in relation to risk assessment and resource-planning. This asymmetry between negative and positive outlier detection persists across different threshold and validation setups in case of different models. The test regarding the importance and relative contribution of feature groups also underlines and validates the initial research purpose and practical business relevance: physiological signals have the biggest (88% in RF) contribution towards predictive power, while self-reports and scene metadata add relatively less in the context of this research using the Emognition dataset. This remains the strongest argument for the introduction of the pipeline and the usage of physiological data as complementary to existing practices at test screenings. Another important finding is that the analysis reached prediction ceiling using the Emognition dataset available. This is a cross-method convergence which proves that the ceiling

is set by the number of observations rather than methodological ceiling. While it questions the validity of the exact results, it also serves as an argument for scalability and potential regarding the pipeline that was designed to use different models with the goal of extracting as many different signals as possible.

With these results, the general question of whether physiological reaction measurement through a scalable pipeline can serve as a complement to self-report surveys in film test-screening scenarios is provided with a qualified and affirmative answer. The claim remains that the pipeline does not replace current practices but rather augment them.

Therefore, this thesis contributes on a methodological level by designing and building the reproducible pipeline, and on an empirical level by applying it to the Emognition dataset, acquiring useful business information, capturing signals and proposing possible use cases in a real-life environment.

Due to the limitations set by data, resources and research scope, there is significant room for further research that must be acknowledged. The biggest current limitation of research is the low number of observations in the Emognition dataset ($n=10$), creating a bottleneck where analysis ends up reaching a prediction ceiling across the whole pipeline. If the pipeline could be tested with data collected via the same method performed on a bigger audience and perhaps using full-length feature films as stimuli, the patterns and signals could be validated further, and more insights were to be gained. To put it in simple terms, current research could easily be elevated by obtaining more data even while keeping the whole pipeline intact.

The current pipeline uses the vast majority of features and channels that are used by the Emognition dataset while capturing the physiological reactions to film clips. However, lots of generalization is done by computing means and standard deviations in favor of efficiency. This representation of the data neglects the time series nature of a film viewing experience and the progression of engagement and arousal dynamics over time. Developing a more advanced convolutional neural network architecture to deal with the raw data would make it possible to test whether these tendencies and patterns carry additional predictive value. The pipeline could also be tested in a way where movie-level aggregation is not performed and individual, subjective perspectives of participants are examined. Naturally, that would also require much more data to be sufficient and for the results to be significant.

Additionally, the introduction of more complex outcome variables to represent perception and profitability would bring more validity and most likely better results. The current thesis had an emphasis on practical business implications and efficiency, but by diving deeper into consumer psychology, movie industry trends across time and marketing theory, it would be possible to find better target variables.

Further research could also include comparative studies with already existing literature. This is partially done in this thesis in case of the closest works relating to this research, but the thorough examination of whether the findings are in alignment with non-physiological movie success prediction literature is outside the scope of this project.

Lastly, the most ideal continuation of this research would definitely be the direct application of it in a test screening scenario. Using the same data collection methods and having a population-representative group of participants watch whole movies would create room for a more reliable critique of pipeline capabilities in signal extraction and model results.

In conclusion, the thesis achieved results which support the proposition that physiological data could serve as complementary to self-reports in test screening scenarios. With explicit limitations, the thesis contributes empirically by providing direct insights about the relationship between physiology, perception and profitability and also methodologically by developing a pipeline of data processing and analysis with focus on deployability, scalability and efficiency.

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Appendices

Appendix 1: Thesis data documentation

Appendix 2: Thesis data in a compressed format (split to ML/DL in a pipeline-process model-ready format)

Appendix 3: Code for feature extraction and data simulation for both ML and DL.

Appendix 4: Code for all analysis across regression, classification and deep learning

Appendix 5: Code for all results for the thesis

Appendix 6: Declaration of use of GAI document